

Technical report — Apertus QA: grounded Spanish answers over official statistics

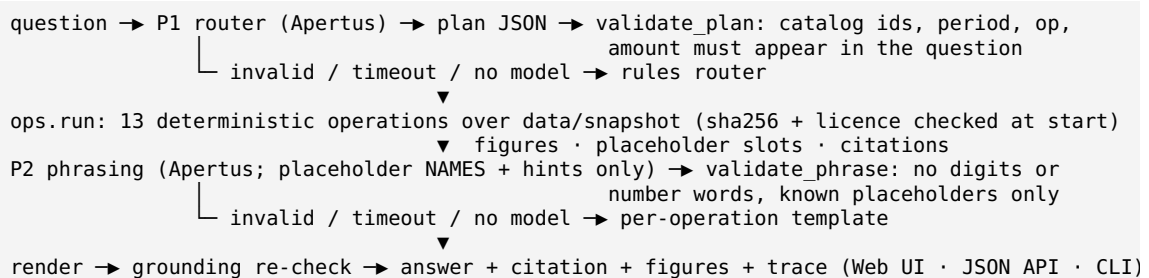
- **Track:** Track 2B — Apertus QA (own project)
- **Event:** Online
- **Team:** Vento Labs — `[[ENTRANT_NAME]]`
- **Demo:** `[[VIDEO_URL]]` (≤ 2 min, screen capture + captions)
- **Code:** `[[REPO_URL]]` @ `[[COMMIT]]`

Draft status. Numbers marked *STUB* come from replayed outputs written by the build agent, not by Apertus, or from the no-model baseline. `[[REAL_RESULTS: ...]]` marks space reserved for real Apertus runs (≈ 0.3 – 0.5 page).

1. Summary

General-purpose chatbots answer statistical questions fluently but often with unsourced, stale or wrong figures. For a newsroom, public office or school that is worse than no answer. **Apertus QA** answers Spanish questions over a local, licence-checked snapshot of official Argentine open data (47 series and tables from INDEC and other agencies, CC BY 4.0). Apertus does the language work: it **plans** the query (series, operation, period) and **phrases** the answer. It never produces or sees a number. Deterministic code computes every figure and attaches the citation. The system says when the data cannot answer, and refuses forecasts and financial advice. Everything except the model call runs with `docker run --network none`, and the model can be a self-hosted Apertus on the same isolated network. *STUB/baseline*: main set 31/31, held-out paraphrases 8/13, 0 grounding violations, 5/5 INDEC cross-checks. `[[REAL_RESULTS: one-sentence headline — Apertus 8B main x/31, held-out y/13, grounding violations z]]`

2. Architecture



One container, one stateless process: Python 3.12 **standard library only** (no pip install, no database). The ≈ 0.3 MB snapshot ships in the image. The Spanish UI is one HTML file with no external assets (a test asserts it). Modules (`src/apertus_qa/`): `catalog`, `ops` (value, latest, change, monthly, year-on-year, max, min, annual total, ranking, inflation adjustment, comparison, table top-n/value), `gateway`, `providers` (OpenAI-compatible, stub replay, recorder), `prompts`, `validate`, `rules`, `pipeline`, `server`. **LLM gateway** (ported from the team's *Ciclo* prototype, §9): never throws; per-minute/per-day budgets, timeout, circuit breaker (transport failures only), tolerant JSON parsing, schema + semantic validation, deterministic fallback with a machine-readable reason (`network_error`, `http_error`, `bad_response`, `budget_exhausted`, `breaker_open`, validator rejection) shown in the trace and the UI.

Grounding guarantees (each covered by tests):

#	Guarantee	Mechanism
G1	Numbers come only from code	ops.py computes from the verified snapshot
G2	The phrasing model never sees a figure	Gets placeholder names and hints (dirección: subió); a test with a fake endpoint asserts the figure is absent from the request
G3	Model text cannot carry numbers	Any digit or number word → rejected → template
G4	The router cannot invent inputs	Series must exist in the catalog; periods well-formed; an amount to adjust must appear in the question
G5	Final text is re-checked	Every numeric token must equal a computed slot value
G6	Citations are written by code	Publisher, series id, period, licence, portal

Residual failure mode: a valid but *wrong* plan gives a real, cited figure for a slightly different question (§6). The UI shows series and period next to every answer so users can spot it.

Target architecture (mandatory): b) Air-gapped — also a) on-premise and c) sovereign Swiss cloud

Stage	External network	How
Build	Base image only (python:3.12-slim)	All RUN steps run under docker build --network none. Isolated sites: make image-save → docker load.
Data	None	Snapshot inside the image; sha256 + licence checked at start; refreshed outside the site as a new image.
Runtime: UI, API, computation, citations	None	Proven with --network none (below).
Runtime: model	Only LLM_BASE_URL	CSCS-hosted Apertus (Switzerland) or self-hosted (vLLM/Ollama/llama.cpp, OpenAI-style API, key optional) in the same isolated network.
Model unreachable	None	Rules + templates: answers stay correct and cited; the UI shows the mode.

Air-gap proof (make offline-proof; full transcript in docs/OFFLINE.md). Under --network none only lo exists; connections to 1.1.1.1:443 / 8.8.8.8:53 fail (Network is unreachable) and DNS fails; the server answers /health and /api/ask inside the container; 35 tests and all four eval runs pass with unchanged scores; in real mode the trace shows fallback (network_error) and the answer is still correct and cited — the model call is the only network dependency. docs/deploy/docker-compose.airgap.yml sketches app + vLLM with local Apertus weights on an internal: true network (not run: no GPU).

3. Use of Apertus

Aspect	Setup
Model	swiss-ai/Apertus-v1.5-8B [[REAL_RESULTS: confirm exact LLM_NAME; 70B if run]]
Usage	Inference in two roles: P1 tool-use-style planning (one JSON plan over a 47-entry catalog) and P2 constrained generation (Spanish phrasing with placeholders only)
Serving	CSCS endpoint or any OpenAI-style server via LLM_BASE_URL, LLM_NAME, LLM_API_KEY; /chat/completions, temperature 0; optional response_format: json_object
P1 prompt	Spanish instructions + compact catalog (ids, titles, units, frequency, coverage; no values ; ≈4k tokens) → {action, op, series, period, from, to, amount, n, ...} or {action: "refuse", reason} (out of scope / forecast / advice / other). Out-of-coverage periods pass through so code can explain coverage.
P2 prompt	Question, operation, placeholder descriptions, direction hints → {"texto": ...}, one or two sentences
Record/replay	LLM_MODE=record saves real outputs, so judges can replay a real run with no key or network (labelled as replay)

[[REAL_RESULTS: observed behaviour — JSON validity rate, typical plan errors, phrasing rejections by reason, latency p50/p95, prompt iterations]]

No other model is used at runtime or in evaluation (scoring is deterministic code, no LLM judge).

4. Data

Domain	Content	Publisher
Prices, wages	CPI national (monthly, Dec 2016 → Aug 2026), Dec–Dec inflation; registered wage index	INDEC
Population, births	Population 2004–2025; live births (to 2022, declared)	INDEC; DEIS / Min. Salud
Exchange rate, trade	Nominal annual exchange rate; exports, total + 4 categories	INDEC / Min. Economía
Tourism	Resident tourist trips by destination and mode (27 series)	Subsecretaría de Turismo
Energy	Installed solar / wind / total capacity; crude oil production	CAMMESA; Secretaría de Energía
Transport, road safety	Subway passengers; 2025 flights by airport and route (counts only); road deaths	INDEC; ANAC; SNIC

All data are **third-party open data under CC BY 4.0** from `datos.gob.ar / apis.datos.gob.ar`, fetched at build time only, respecting `robots.txt` and its crawl delay. We redistribute them under their original licence **with attribution** (`data/README.md`, `docs/DATA.md`, and a citation on every answer); we do not relicense them or publish them as our own dataset. Transformations: time-window trimming; flight and road-death microdata reduced to aggregate counts; no personal data. Per-file sha256 in `catalog.json`.

5. Evaluation

Sets. *Main*: 31 Spanish questions — 23 answerable, 8 to refuse (missing period, forecast, out of scope, advice, prompt injection). *Held-out*: 13 paraphrases (colloquial, elliptical) written separately and not used to build the rules router or prompts. **Ground truth** is computed from the snapshot by independent code (`src/eval/make_expected.py`, which does not import the app); five CPI answers are also checked against INDEC's published figures. **Metrics**: decision (answer vs refuse), value match, citation (series + period), grounding violations (numbers not traceable to a computed figure; target 0), false refusals, fallback rates by reason.

Setup	Set	Overall	Answers	Refusals	False refusals	Grounding viol.
No-model baseline (rules + templates)	main	31/31	23/23	8/8	0	0
STUB replay (agent-written, not Apertus)	main	31/31	23/23	8/8	0	0
No-model baseline	held-out	8/13	6/11	2/2	4	0
Apertus 8B	main	[[REAL_RESULTS]]				
Apertus 8B	held-out	[[REAL_RESULTS]]				

All rows reproduced in the Docker image with `--network none`; INDEC cross-checks 5/5. **Reading.** The rules router was written while looking at the main set, so its 31/31 is not blind; the held-out set is the fair comparison. Claim under test: *Apertus raises held-out coverage above the baseline's 8/13 while keeping grounding violations at 0*. STUB rows only show that pipeline, validators and fallbacks work; its four adversarial outputs (unknown series id, markdown-fenced plan, invented figure, banned word "récord") are all handled as designed.

[[REAL_RESULTS: error analysis ≈ 0.3 page — per-category breakdown; plan errors caught vs passed; phrasing rejections; optional 70B row]]

6. Limitations

Fixed coverage (47 entries; no unemployment, poverty or GDP yet; CPI to Aug 2026). A valid but wrong plan can answer a slightly different question (1/13 held-out for the baseline); plan-question consistency checks are planned. Spanish only, single turn. Real-model results pending; self-hosted Apertus untested (no GPU); no load or accessibility testing.

7. Reproducibility

From `track_2b/`: `make run` (UI at `http://localhost:8080`), `make test`, `make eval`, `make eval-heldout`, `make offline-proof`, `make report` (this PDF). STUB and off modes are deterministic: `docs/eval/` regenerates identically (apart from a timing field). Verified on Docker Engine 29.8.2, Linux x86-64, CPU only. Without Docker: `make run-local / test-local / eval-local` (Python ≥ 3.10). Real runs: set `LLM_BASE_URL`, `LLM_NAME`, `LLM_API_KEY`, then `LLM_MODE=record make eval`. Commit: `[[COMMIT]]`.

8. Next steps

Real Apertus runs (8B, then 70B) with prompt iteration on the held-out set; plan-question consistency checks; more series (labour market, GDP, provinces); a second statistics office (e.g. BFS / `opendata.swiss`) through the same catalog format; a measured self-hosted deployment on an isolated GPU node.

9. Disclosures

Topic	Statement
AI-assisted development	Built with closed-weights (proprietary) AI coding agents , directed and reviewed by the entrant. They wrote the code, tests, docs and this report. They are not part of the submitted system, are not called at runtime, and are not used as judges. The only runtime model is Apertus. (The rules mention open-weights models supporting development; ours were closed-weights, so we state their role explicitly.)
STUB outputs	Until real runs are recorded, the STUB replays outputs written by the build agent , labelled STUB in the UI, eval reports and this report.
Component reuse	LLM gateway ported (JavaScript $\rightarrow$ Python) from the team's Ciclo prototype, written 2 Oct 2026; one change: invalid content no longer trips the breaker. Container/ <code>env-var/health</code> pattern adapted from Ciclo.
Data reuse	Raw downloads, licence evidence and aggregation definitions from the team's open-data pipeline (1–2 Oct 2026), re-verified at build time.
Template	HackApertus/project-template @ 7f23822 (Apache-2.0). Details: <code>docs/DISCLOSURE.md</code> .
Data licence	Third-party CC BY 4.0 with attribution (§4).

License

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References

Apertus v1.5 model card (`huggingface.co/swiss-ai/Apertus-v1.5-8B`) · Series de Tiempo API, `datos.gob.ar` (`datosgobar.github.io/series-tiempo-ar-api`) · INDEC, Índice de precios al consumidor (`indec.gob.ar`) · Hack Apertus project template (`github.com/HackApertus/project-template @ 7f23822`)